STA 630 final 2021

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1. **Historical comparative research limitation**
2. **Content analysis**
3. **Simple random sampling and stratified random sampling**

   Simple Random Sample

   The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

   Stratified Random Sample

   When the population is heterogeneous, the use of simple random sample may not produce representative sample. Some of the bigger strata may get over representation while some of the small ones may entirely be eliminated. Look at the variables that are likely to affect the results, and stratify the population in such a way that each stratum becomes homogeneous group within itself. Then draw the required sample by using the table of random numbers. Hence in stratified random sampling a sub-sample is drawn utilizing simple random sampling within each stratum. (Randomization is not done for quota sampling).
4. **Why we prefer similar population?**
5. **Why editing is required by field worked during data collection process?**
6. **Experimental research is a qualitative research which build on interpretivists principles agree or not agree?**
7. **Researchers who conduct research on deviant, engaged in illegal behavior face an additional dilemma mentioned the dilemma for the field researcher?**
8. **How we make questions for field work research**

   Data collection in a sponsored study is rarely carried out by the person who designs the research project. For a student, depending upon the sample size, data collection is usually done by the student himself/herself. However, the data collection stage is crucial, because the research project is no better than the data collected in the field. Therefore, it is important that the research administrator selects capable people who may be entrusted to collect the data. There are Field Interviewing Services, who specialize in data gathering. These agencies perform door-to-door surveys, central location telephone interviewing, and other forms of fieldwork for fee.

9. **Double blind experiment**

   The double-blind experiment is designed to control experimenter expectancy. In it, people who have direct contact with subjects do not know the details of the hypothesis or the treatment. It is double blind because both the subjects and those in contact with them are blind to details of the experiment. For example a researcher wants to see if new drug is effective.
10. **Maturity effect and test effect sampling techniques**

Cause and effect relationship can also be contaminated by the effects of the passage of time – another uncontrollable variable. Such contamination is called maturation effect. The maturation effects are a function of the processes – biological and psychological – operating within the subjects as a result of the passage of time. Examples of maturation processes could include growing older, getting tired, feeling hungry, and getting bored. In other words there could be maturation effect on the dependent variable purely because of the passage of time

11. **Parts of the table**
    - Give each table a number.
    - Give each table a title, which names variables and provides background information
    - Label the row and columns variables and give name to each of the variable categories.
    - Include the totals of the columns and rows. These are called marginal. They equal the univariate frequency distribution for the variable.
    - Each number or place that corresponds to the intersection of a category for each variable is a cell of a table.
    - The numbers with the labeled variable categories and the totals are called the body of the table.
    - If there is missing information, report the number of missing cases near the table to account for all original cases.
    - Researchers convert raw count tables into percentages to see bivariate relationship. There are three ways to percentage a table: by row, by column, and for the total. The first two are often used and show relationship.
12. **Probability or Non-probability**

    Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizability. When time or other factors, rather than generalizability become critical then non-probability sampling is generally used.

    In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment.
13. **Pretest posttest**

    Frequently a researcher measures the dependent variable more than once during an experiment. The pretest is the measurement of the dependent variable prior to the introduction of the treatment. The posttest is the measurement of the dependent variable after the treatment has been introduced into the experimental situation.
14. **Why we use Sample population researchers.**
15. **Random sampling**

    Researchers often use random sampling in content analysis. First, they define the population and the sampling element. For example, the population might be all words, all sentences, all paragraphs, or all articles in certain type of documents over a period of specified period. Likewise, it could include each conversation, situation, scene, or episode of a certain type of television program over a specified time period.
16. **Experiment research**
    a. Experimental research builds on the principles of positivist approach more

directly than do the other research techniques.

b. The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated.

c. The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.

d. Experimentation is a research design that allows evaluation of causal relationship among variables.

17. **How construct frequency distribution in two variable.**
18. **Difference in Internal or external validity**

Internal validity:

Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalizability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situations.

External Validity:

Even if the researcher eliminates all concerns for internal validity, external validity remains a potential problem. External validity is the ability to generalize experimental findings to real life situations. Without external validity, the findings are of little use for both basic and applied research i.e. we shall not be able to develop any theories that could be applicable to similar other situations.

19. **Ethical dilemmas**

The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has little time to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Let us look at some of these dilemmas:

Deception:

Deception arises in several ways in field research. The research may be covert; or may assume a false role, name, or identity; or may mislead members in some way.

Confidentiality:

A researcher learns intimate knowledge that is given in confidence. He or she has a moral obligation to uphold the confidentiality of data. This includes keeping information confidential from others in the field and disguising members‟ names in field notes.

Involvement with deviants:

Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal activity. They might be getting „guilty knowledge.‟ such knowledge is of interest not only to law enforcement officials but also to other deviants.

The powerful:

Field researchers tend to study those without power in society (e.g., street people, the poor, children, and lower level workers). Powerful elites can block access and have effective

gatekeepers. Researchers are criticized for ignoring the powerful, and they are also criticized by the powerful for being biased toward the less powerful.

Publishing field reports:

The intimate knowledge that a researcher obtains and reports creates a dilemma between the right of privacy and the right to know. A researcher does not publicize member secrets, violate privacy, or harm reputations.

20. **Difference b/w simple and systematic random sample**

Simple Random Sample

The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

Systematic Random Sample

Systematic random sampling is simple random sampling with a short cut for random selection. Again, the first step is to number each element in the sampling frame. Instead of using a list of random numbers, researcher calculates a sampling interval, and the interval becomes his or her own quasi random selection method. The sampling interval (i.e. 1 in K where K is some number) tells the researcher how to select elements from a sampling frame by skipping elements in the frame before one for the sample.

21. **Two basic rules for coding construction.**

There are two basic rules for code construction. First, the coding categories should be exhaustive – that is, coding categories should be provided for all subjects or objects or responses. With a categorical variable such as sex, making categories exhaustive is not a problem. However, when the response represents a small number of subjects or when the responses might be categorized in a class not typically found, there may be a problem.
Second, the coding categories should also be mutually exclusive and independent. This means that there should be no overlap between the categories, to ensure that a subject or response can be placed in only one category. This frequently requires that an “other” code category be included, so that the categories are all inclusive and mutually exclusive

22. **Importance of instrumentation in internal validity**

Instrumentation effects are yet another source of threat to internal validity. These might arise because of a change in the measuring instrument between pretest and posttest, and not because of the instrument’s differential impact at the end. For example, in a weightless experiment, the springs on the scale weaken during the experiment, giving lower readings in the posttest. A change in the wording of questions (may be done to avoid testing effects), change in interviewers, or change in other procedures to measure the dependent variable can cause instrumentation effect. Performance of the subjects measured by the units of output in the pretest, but when measuring the output in posttest the researcher measures it by the number of units rejected, and the amount of resources expended to produce the units.

23. **What type of research to be used for economic study of Pakistan?**
24. **Explain the role of pilot testing in development of research instruments.**

It is important to pilot test the instrument to ensure that the questions are understood by the respondents and there are no problems with the wording or measurement. Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their

comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled. Making a mistake with 25 or so subjects can avert the disaster of administering an invalid questionnaire to several hundred individuals. Hence the main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

25. **A researcher takes data of 15 persons first. And then again take interview of 5 persons among these 15.. What type of sampling is used?**
26. **Pilot testing long**

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire.

Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled.

What aspects to be evaluated during pilot testing?

i. **Reactions of Respondents:**

The reactions of the respondents can be looked at from different angles. The researcher may be familiar with the local culture; still getting the first-hand experience is always useful. Going to the field, contacting the people, and their reactions to the different aspects of research may be a learning experience.

- Acceptability of the questions asked.
- Willingness of the respondents to co-operate.

ii. Discovering errors in the instrument:
- Do the tools provide you the information? Reliability. Suitability for analysis
- Time taken/needed to interview/conduct the observation.
- If there is any need to revise the format of the tool

iii. Sampling procedure can be checked:
- The extent to which instructions given are followed
- How much time is needed to locate the respondents?

iv. Staffing and activities of research team can be checked:
- How successful the training has been?
- What is the work output of each member?
- How well the research team works together?
- Is the logistical support adequate?

v. Procedure for data processing and analysis can be evaluated:
- Make dummy tables. S

27. **FGD long**

In research, focus group discussions (FGD) are a more formal way of getting groups of people to discuss selected issues. A focus group discussion is a group discussion of 6-12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic. There may be some disagreement about the exact number of participants in the discussion, as one comes across variations in numbers (6 to10, 6 to12, 6 to15, 8 to10, 5 to 7) in different books on research methods.

Uses of Focus Group Discussions

- The primary advantage of focus groups is its ability to quickly and inexpensively grasp the core issues of the topic.
- As part of exploratory research, focus group discussions help the researcher to focus on the issue and develop relevant research hypotheses.
- Focus group discussion is an excellent design to get information from non-literates.
- Focus groups discussions are a good means to discover attitudes and opinions that might not be revealed through surveys.
- Focus group discussions are well accepted in the folk communities, as this form of communication already exists whereby the local communities try to sort out controversial issues.
- Focus group discussions generate new ideas, questions about the issues under consideration. It may be called serendipity (surprise ideas). It is more often the case in a group than in an individual interview that some idea will drop out of the blue.
- Focus group discussions can supplement the quantitative information on community knowledge, attitude, and practice (KAP), which may have already been collected through survey research.
- Focus group discussions are highly flexible with respect to topic, number of participants, time schedule, location, and logistics of discussion.
- Focus group discussions provide a direct link between the researcher and the population under study.
- For some researchers, focus group discussions may be a fun. They enjoy discussing the issues directly with the relevant population.

28. Field research

Observation studies can be participant or non-participant. In participant observation the researcher directly observes and participates in small scale social settings in the present time. Such a study is also referred to as **field research**, ethnography, or anthropological study. Here the researchers:

- Study people in their natural settings, or in situ.
- Study people by directly interacting with them.
- Gain an understanding of the social world and make theoretical statements about members‟ perspective.

Field researchers study people in a location or setting. It has been used to study entire communities. Field research has a distinct set of methodologies. Field researchers directly observe and interact with community members in natural settings to get inside their perspectives

29. Two situations

Situation 1: when you are going to a company and ask them to talk about the features of the mobile you used r like...

Situation 2: when you are visiting the automobile survey and assessing its methods...

Tell about both situations which type of research type is used in it?

30. Non-reactive research

Experiments and surveys research are both reactive; that is, the people being studied are aware of the fact that they are being studied. In non-reactive research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

31. Specification Pattern

Is the next easiest pattern. It occurs when one partial replicate the initial bivariate relationship but other partials do not.

The interpretation pattern

Describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

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Mostly paper conceptual

CONDUCTS THE FIELDWORK

Data collection in a sponsored study is rarely carried out by the person who designs the research project. For a student, depending upon the sample size, data collection is usually done by the student himself/herself. However, the data collection stage is crucial, because the research project is no better than the data collected in the field. Therefore, it is important that the research administrator selects capable people who may be entrusted to collect the data. There are Field Interviewing Services, who specialize in data gathering. These agencies perform door-to-door surveys, central location telephone interviewing, and other forms of fieldwork for fee.

NON-REACTIVE RESEARCH

Experiments and surveys research are both reactive; that is, the people being studied are aware of the fact that they are being studied. In non-reactive research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

PROBABILITY AND NON-PROBABILITY SAMPLING

There are several alternative ways of taking a sample. The major alternative sampling plans may be grouped into probability techniques and non-probability techniques. In probability sampling every element in the population has a *known nonzero probability* of selection. The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizability. When time or other factors, rather than generalizability become critical then non-probability sampling is generally used. In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate.

Types of non-probability sampling

In non-probability sampling designs, the elements in the population do not have any probabilities attached to their being chosen as sample subjects. This means that the findings from the study of the sample cannot be confidently generalized to the population. Sometimes non-probability could be the only way to collect the data.

1. Convenience Sampling

2. Purposive Sampling

3. Quota Sampling
4. Snowball Sampling
5. Sequential Sampling
6. Theoretical Sampling.

The replication pattern is the easiest to understand. It is when the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable. It means that the control variable has no effect.

The specification pattern

Is the next easiest pattern. It occurs when one partial replicate the initial bivariate relationship but other partials do not.

The interpretation pattern

Describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

EXPERIMENTAL RESEARCH

- Experimental research builds on the principles of positivist approach more directly than do the other research techniques.
- The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated.
- The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.
- Experimentation is a research design that allows evaluation of causal relationship among variables.

TYPES OF PROBABILITY SAMPLING and TYPES

Probability samples that rely on random processes require more work than nonrandom ones. A researcher must identify specific sampling elements (e.g. persons) to include in the sample. For example, if conducting a telephone survey, the researcher needs to try to reach the specific sampled person, by calling back several times, to get an accurate sample. Random samples are most likely to yield a sample that truly represents the population. In addition, random sampling lets a researcher statistically calculate the relationship between the sample and the population – that is the size of *sampling error.* A non-statistical definition of the sampling error is the deviation between sample result and a population parameter due to random process.

1. Simple Random Sample:
2. Systematic Random Sample:
3. Stratified Random Sample:
4. Multistage Sampling:
5. Cluster Sampling:
6. Double Sampling:

Cluster Sampling

- Purpose: sample economically and retaining probability sampling.

- □ Heterogeneity within clusters but homogeneity between clusters.
- □ Random selection of clusters.
- □ Random selection of elements within the selected clusters.
- □ Cluster sampling addresses two problems
- □ Researchers lack good sampling frame for dispersed population
- □ Reaching each sampled element is costly.

Coding

- Coding involves assigning numbers or other symbols –responses can be grouped into classes or categories
- Helps in efficient analysis of data.
- Keep the data in raw form so far it is possible. Computer can help in grouping.
- ABU SULTAN's SHORT-NOTES – RESEARCH METHODS (STA630) – LECTURE#23 to 37--------Page 18 of 34
- Codes allow data processing in a computer.
- Organize data into field, record and file.
- A field: collection of characters (single No., letter, symbol) representing single type of data.
- A record: collection of related fields.
- A file: collection of related records

Validity in Experiments

Experiments are judged by two measures; internal validity and external validity. The first, internal validity indicates whether the independent variable was the sole cause of the change in the dependent variable. It implies the researcher's ability to eliminate alternative explanations of the dependent variable. Variables, other than the treatment, that affect the dependent variable are threats to internal validity. They threaten the researcher's ability to say that the treatment was the true causal factor producing change in the dependent variable. The second measure, external validity, indicates the extent to which the results of the experiment are applicable in the real world. Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalizability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situations. Factors Affecting Internal Validity: There are eight major types of extraneous variables that may jeopardize internal validity:

1. History effect
2. Maturation effect
3. Testing effect
4. Instrumentation effect
5. Selection bias effect
6. Selection bias effect
7. Statistical regression
8. Mortality
9. Mechanical loss

INDEX CONSTRUCTION

An index is combination of items into a single numerical score. Various components or subgroups of a construct are each measured and then combined into one measure. There are many types of indexes. For example, if you take an exam with 25 questions, the total number of questions correct is a kind of index. It is a composite measure in which each question measures a small piece of knowledge, and all the questions scored correct or incorrect are totaled to produce a single measure. One way to demonstrate that indexes are not a very complicated is to use one. Answer yes or no to the seven questions that follow on the characteristics of an occupation.

PILOT TESTING OF THE Questionnaire

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire. Conventional wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for extra training the research team.

Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. The main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

WHAT ASPECTS TO BE EVALUATED DURING PILOT TESTING

1. Reactions of Respondents:
2. Discovering errors in the instrument:
3. Sampling procedure can be checked:
4. Staffing and activities of research team can be checked:
5. Procedure for data processing and analysis can be evaluated

Systematic Random Sample

- Short cut for simple random sampling.
- Number each element in the sampling frame.
- Instead of using the table of random numbers, researcher develops a system for the selection of elements.
- Calculate the sampling interval. Basis of random selection

Probing

Probing means the verbal prompts made by field worker when the respondent must be motivated to communicate his or her answer or to enlarge on, clarify or explain an answer. Probing may be needed for two types of situations. First, it is necessary when the respondent must be motivated to enlarge on, clarify, or explain his or her answer.

Snowball Sampling

Snowball sampling (also called network, chain referral, or reputational sampling) is a method for identifying and sampling (or selecting) cases in the network. It is based on an analogy to a snowball, which begins small but becomes larger as it is rolled on wet snow and picks up additional snow. It begins with one or a few people or cases and spreads out on the basis of links to the initial cases. This design has been found quite useful where respondents are difficult to

identify and are best located through referral networks. In the initial stage of snowball sampling, individuals are discovered and may or may not be selected through probability methods.

Reliability and Validity

Validity problems occur when the researcher's theoretical definition does not match that of the government agency or organization that collected the information. Official policies and procedures specify definitions for official statistics.

Reliability

Stability reliability problems develop when official definition or the method of collecting information changes over time. Official definitions of work injury, disability, unemployment, literacy, poverty, and the like change periodically. Even if the researcher learns of such changes, consistent measurement over time is impossible. Equivalence reliability can also be a problem.

Secondary Data

Secondary data refer to information gathered by someone other than the researcher conducting the present study. Secondary data are usually historical, already assembled and do not require access to respondents or subjects.

Factors Affecting Internal Validity

There are eight major types of extraneous variables that may jeopardize internal validity:
1. History effect
2. Maturation effect
3. Testing effect
4. Instrumentation effect
5. Selection bias effect
6. Selection bias effect
7. Statistical regression
8. Mortality
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Types of Designs: Researchers combine parts of experiment (e.g. pretests, control groups, etc.) together into an experimental design. For example some designs lack pretests, some do not have control groups, and others have many experimental groups. Certain widely used standard designs have names. Classical Experimental Design: All designs are variations of the classical experimental design, which has random assignment of subjects, a pretest and a posttest, an experimental group, and a control group.
1. Quasi Experimental Designs
2. True Experimental Designs
3. Solomon's Four Group Design

Quasi-Experimental Designs

One-shot Case Study Design

Also called the one-group posttest-only design.
The one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment.

One Group Pretest-posttest Design

This design has one group, a pretest, a treatment, and a posttest. It lacks a control group and random assignment.

Two Groups Posttest-only Design:

It has two groups, a random assignment of subjects, a posttest, and a treatment. It has all parts of the classical design except a pretest.

True Experimental Designs

Experimental designs, which have at least two groups, a random assignment of subjects to experimental and control groups, only experimental group is exposed to treatment, both groups record information before and after the treatment, are known as ex-post facto experimental designs.

Random Assignment

Random assignment is a method for assignment cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way.

Parts of Experiments

We can divide the experiments into seven parts and for each part there is a term. Not all experiments have all these parts, and some have all seven parts plus others. The following seven usually make up a true experiment. 1. Treatment or independent variable. 2. Dependent variable. 3. Pretest. 4. Posttest. 5. Experimental group. 6. Control group. 7. Assignment of subjects.

BIVARIATE TABLES

The bivariate contingency table is widely used. The table is based on cross-tabulation (cross classification); that is the cases are organized in the table on the basis of two variables at the same time. A contingency table is formed by cross-tabulating the two or more variables.

PRINCIPLES OF INTERVIEWING —5

Have integrity and be honest. This is the cornerstone of all professional inquiry, regardless of its purpose.

1 Have patience and tact. Interviewers ask for information from people they do not know. Thus all the rules of human relations that apply to inquiry situations – patience, tact, courtesy – apply "in spades" to interviewing.

2 Have attention to accuracy and detail. Among the greatest interviewing "sins" are inaccuracy and superficiality, for the professional analyst can misunderstand, and in turn mislead, a client. Do not record the answer unless you fully understand it yourself. Probe for clarification and detailed full answers.

3 Exhibit a real interest in the inquiry at hand, but keep your opinions to yourself. Impartiality is imperative.

4 Be a good listener. Some interviewers talk too much, wasting time when respondents could be supplying more pertinent facts or opinions on the topic.
5 Keep the inquiry and respondents' responses confidential. Do not discuss the studies you are doing with relatives, friends, or associates. Never quote one respondent‟s opinion to another.
6 Respect others' rights. Survey research depends on the goodwill of others to provide information

Steps in Field Research

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. Here is just the listing of these steps.

1. Prepare yourself, read the literature and defocus.

2. Select a site and gain access.

3. Enter the field and establish social relations with members.

4. Adopt a social role, learn the ropes, and get along with members.

5. Watch, listen, and collect quality data.

6. Begin to analyze data, generate and evaluate working hypothesis.

7. Focus on specific aspects of the setting and use theoretical sampling.

8. Conduct field interviews with member informants.

9. Disengage and physically leave the setting.

10. Complete the analysis and write the report.

Quantitative approach

Positivist researchers reject the idea that there is a distinct H-C method. They measure variables, test hypotheses, analyze quantitative data, and replicate research to discover generalizable laws that hold across time and societies. They see no fundamental distinction between quantitative social research and historical-comparative research. They apply quantitative research techniques, with some minor adjustments, to study the past or other cultures

Qualitative approach

There are no easily listed steps to follow in the analysis of historical data. Max Weber used the German term *verstehen* –"understanding" – in reference to an essential quality of research in behavioral sciences. He meant that the researcher must be able to take on, mentally, the circumstances, views, and feelings of those being studied to interpret their actions appropriately. The historical-comparative researcher must find patterns among the voluminous details describing the subject matter of study.

Similarities to Field Research

Stratified Random Sample

When the population is heterogeneous, the use of simple random sample may not produce representative sample. Some of the bigger strata may get over representation while some of the small ones may entirely be eliminated. Look at the variables that are likely to affect the results, and stratify the population in such a way that each stratum becomes homogeneous group within itself. Then draw the required sample by using the table of random numbers. Hence in stratified random sampling a sub-sample is drawn utilizing simple random sampling within each stratum. (Randomization is not done for quota sampling)

TOOLS FOR DATA COLLECTION----5